

# Understanding Attention Patterns toward Embodied AI Agents in Immersive Learning Environments

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**Figure 1: Overview of embodied interaction in ChromosoME: (A) a learner performing physical interaction, (B) the corresponding immersive VR view during manipulation, and (C) an embodied AI agent providing gesture-based guidance.**

## Abstract

We present an exploratory study examining learner interaction with a gesture-performing AI agent embedded in an immersive VR simulation for learning mitosis. Through video-based interaction analysis of 17 participants (74 interaction episodes), we investigated when and how learners attend to embodied AI agents during hands-on VR learning. While participants rated overall system usability highly, AI agent-specific items received relatively lower ratings, revealing tension between static interface elements and adaptive agent support. Interaction patterns showed learners engaged agents primarily during uncertainty or for confirmation, but frequently disengaged when agent guidance overlapped with on-screen text, when dialogue felt structurally constrained, or during response latency. We contribute empirical understanding of attention allocation in agent-supported VR environments and derive design implications for complementary agent roles, gesture timing aligned with learner rhythms, and dialogue architectures supporting interactions while maintaining task guidance.

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## 1 Introduction & Background

Immersive virtual reality (VR) has emerged as a promising medium for learning phenomena that are difficult to convey through static or screen-based representations, particularly in STEM domains where processes unfold dynamically across space and time. VR supports learning by enabling learners to engage with content through bodily action and immersive perception, often increasing engagement and conceptual understanding [17]. However, studies have also documented recurring usability challenges in educational VR systems, including increased cognitive load, disorientation, and difficulties interpreting interface elements [8, 16]. These findings highlight that learning outcomes in VR depend not only on immersion, but also on how learners interact with interface components and manage attention during task execution.

At the same time, AI agents are increasingly integrated into learning environments to provide adaptive support and scaffolding [6, 18]. Prior research in human–AI interaction suggests that agent effectiveness depends not only on the information provided, but also on how learners interpret agent behavior and integrate it into ongoing activity [1, 11]. Recent work indicates that the timing and contextual relevance of agent interventions are critical, as poorly aligned support can interrupt task flow and lead learners to disengage [7]. Much of this research, however, has focused on screen-based or conversational agents.

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Embodied cognition theories posit that learning is grounded in bodily action and sensorimotor experience, motivating the use of embodied agents that communicate through nonverbal cues such as gesture [2, 19]. In immersive VR environments, embodied agents have been proposed as a way to situate instructional support within learners' ongoing interaction and physical engagement [5]. However, in such environments, learners must coordinate attention across physical action, immersive perception, and multiple interface components under heightened cognitive and sensorimotor demands [12]. Despite this promise, the design space of gesture-performing AI agents in VR remains underexplored, and there is limited empirical understanding of how learners actually interpret, attend to, and use such agents during embodied VR learning [5]. In particular, prior work has rarely examined how learners allocate attention to embodied agents, how agent behavior interacts with other interface elements, and when such agents support or hinder embodied learning.

To address this gap, we present an exploratory usability and interaction analysis study of a gesture-performing AI agent embedded in an immersive VR simulation for learning mitosis. Grounded in embodied cognition, this study examines learners' interaction with the interface ecology of the system, focusing on how attention to the AI agent emerges or dissipates during embodied interaction and how these patterns inform agent design. Specifically, we investigate the following research questions: **RQ1.** How do learners interpret and use the interface components of an immersive VR simulation that includes an embedded AI agent? **RQ2.** When and under what conditions do learners attend to or disengage from the AI agent during interaction in an immersive VR learning environment? **RQ3.** Based on RQ1 and RQ2, what design insights can inform the development of gesture-performing AI agents?

## 2 VR Simulation Design

The Virtual Reality (VR) simulation, ChromosoME, was designed to support high school students' understanding of cell division (mitosis) by grounding learning in experiential and embodied learning perspectives. The core design goal is to make learners' hands an active part of the cell division process, fostering presence and learning through gestures directly mapped to biological concepts. Developed for wireless VR headsets (e.g., Oculus Quest), the simulation leverages full hand tracking without external sensors. Learners are immersed inside a human cell and use bodily actions to complete the mitosis process and "heal" a virtual patient, repeating the process three times to support both acclimation to VR and reinforcement of cell division as an ongoing biological mechanism.

ChromosoME presents a structured, immersive representation of mitosis aligned with Multimedia Learning Theory [13] by segmenting content into discrete phases. An Information Panel provides phase-specific descriptions, cellular context, and task instructions. Each phase requires learning-congruent gestures: charging ATP by moving nutrients to the mitochondrion during Interphase, condensing DNA through sustained grasping in Prophase, aligning chromosomes in Metaphase, and separating chromatids with two-handed pulling gestures in Anaphase. Telophase is presented automatically without user interaction.

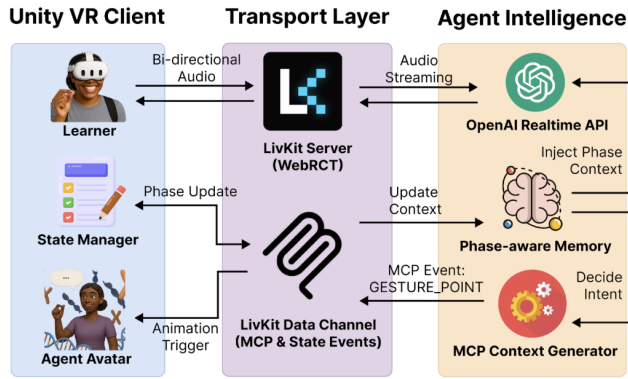
The simulation was refined through an iterative, theory-driven design process. To reduce extraneous cognitive load, the current version of the simulation displays only task-relevant objects, with visual elements intentionally simplified rather than biologically photorealistic. Based on pilot feedback, three design refinements were implemented: (1) the introduction of a tutorial phase that allows learners to practice basic VR interactions prior to the main learning scenario, (2) fixing the position of the Information Panel to reduce distraction and maintain focus on learning content, and (3) expanding interaction tolerances to prevent learners from misinterpreting minor gesture inaccuracies as task failure.

## 3 AI Agent Design

The AI agent in ChromosoME was implemented as a voice-interactive tutoring agent embedded directly within the VR simulation. To support real-time, human-like, and bi-directional audio interaction, the system utilizes the LiveKit Agents framework for Unity as its real-time interface layer. LiveKit Agents manages persistent WebRTC connections between the client and backend, handling audio capture/transmit, audio playback, and interactive text streams while seamlessly integrating with OpenAI's Realtime API as the underlying language and speech model provider. This configuration enables the agent to perform on-the-fly speech-to-text and intent interpretation, generate contextually grounded responses, and output synthesized speech without breaking immersion or inserting conspicuous delays.

At the heart of the agent's interaction logic is a phase-aware dialogue controller that correlates Unity's simulation state with conversational policy. As learners progress through mitosis stages (e.g., prophase, metaphase, and anaphase), the controller updates a state memory service that conditions the agent's responses and guiding prompts. LiveKit Agent sessions forward learner speech to the Realtime API, which uses the current phase context as part of the prompt injection to produce tailored feedback without straying from instructional constraints, ensuring that the agent's utterances remain coherent to the simulation's progression without imposing hard resets on learner agency. Additionally, a dialogue router [20] distinguishes between lightweight clarification intents (e.g., "Did I do this right?") and deeper instructional interventions, enabling responsive yet structured guidance that aligns with real-time learner needs.

To coordinate language output with embodied demonstrations, the system employs Model Context Protocols (MCP) as structured context signals exchanged alongside natural language. MCP messages encode interaction intents, such as `GESTURE_POINT`, `GESTURE_PULL_APART`, or `ATTENTION_REQUEST`, that are emitted by the agent when a gesture or visual cue is pedagogically relevant. These MCP events are transmitted to the Unity client through LiveKit's data channel and interpreted locally to trigger predefined gesture animations. By separating gesture intent from spoken output, the agent avoids embedding action instructions directly into language, allowing gestures to be scheduled, delayed, or suppressed based on learner attention and task engagement.



**Figure 2: System overview of the AI agent pipeline.** Audio is streamed between the Unity client and the OpenAI Realtime API via LiveKit, while phase context is injected through a phase-aware memory module. Gesture behaviors are triggered separately via MCP intent events sent over the data channel.

## 4 User Study

We conducted a usability study to examine how learners interpret the system and interact with an AI agent while learning the process of cell division in an immersive learning environment. After obtaining approval from the Institutional Review Board (IRB), participants were recruited from across a large R1 public university located in the western United States. A total of 18 undergraduate students participated in the study ( $N = 18$ ,  $M \approx 19.93$  years; 13 female, 5 male).

### 4.1 Procedure

The study was conducted in a one-on-one interview format. Upon arrival, participants reviewed and signed an informed consent form, then completed a pre-test assessing key concepts of cell division. Participants were then provided with a one-page handout to study for as long as they wished (most spent 2–3 minutes). Afterward, participants engaged with the ChromosoME VR simulation. During the VR session, the researcher recorded both (a) an external video capturing the participant’s physical interaction with the VR system (Fig.1, left) and (b) a screen recording of the participant’s immersive VR view (Fig.1, center). These recordings enabled analysis of how participants interacted with the system and when they visually attended to the AI agent. After completing the VR simulation, participants filled out a post-survey using a 5-point Likert scale to assess usability, focusing on ease of use, perceived usefulness, and immersiveness. Open-ended feedback was also collected. Finally, participants completed a post-test with the same items as the pre-test, with question order randomized.

### 4.2 Coding Scheme

We developed a video-based coding scheme to analyze learners’ interaction with the AI agent during VR learning. Each interaction episode was coded by initiation source (researcher-prompted, agent-initiated, or self-initiated) and visual attention to the agent

(immediate, delayed, or absent), and combined into composite codes. Full definitions and examples are provided in the Appendix.

## 5 Results

### 5.1 Overall Usability Results

We administered a post-survey to assess perceived ease of use, usefulness, and immersion of the VR simulation with an embedded gesture-performing AI agent (see Appendix for the full post-survey items and results). Participants reported generally high ratings across all dimensions, with ease of use rated highest ( $M = 4.23$ ,  $SD = 0.49$ ), followed by usefulness ( $M = 4.06$ ,  $SD = 0.58$ ) and immersiveness ( $M = 3.98$ ,  $SD = 0.92$ ). These results indicate that participants could readily understand and operate the system and generally perceived it as supportive for learning mitosis. However, items related to AI agent interaction (Items 10, 11, 16, 17, and 20) received relatively lower and more variable ratings. To better understand learners’ engagement with the AI agent in context, we conducted a detailed video interaction analysis.

### 5.2 Learning Outcomes

To assess whether the VR simulation plausibly supported learning, participants completed a short pre-test and post-test before and after the VR session. On average, post-test scores were higher than pre-test scores across participants (Pre:  $M = 3.39$ ,  $SD = 1.65$ ; Post:  $M = 5.06$ ,  $SD = 1.11$ ), indicating improvement in conceptual understanding of mitosis following interaction with the simulation. These results are included to contextualize the interaction analysis by demonstrating that learning took place. As the primary aim of this study is to investigate learner–agent interaction rather than to evaluate learning outcomes in depth, we focus the remainder of the Results section on interaction dynamics.

### 5.3 Interaction Patterns

We conducted a video-based interaction analysis of 17 participants to examine when and how learners engaged with the embedded AI agent during embodied task execution; one participant was excluded due to corrupted video data. Across the dataset, we identified 74 interaction episodes. Of these episodes, 18 were researcher-prompted (A1), 32 were agent-initiated (A2), and 24 were self-initiated (A3). Visual attention to the AI agent was observed in 64 episodes.

We observed four recurring engagement patterns that reflected learners’ motivations for attending to the AI agent:

- **Help-Seeking during Uncertainty.** During moments of procedural or conceptual uncertainty, participants turned to the agent to seek guidance, such as when they were unsure about the next step or the meaning of a particular phase. In these cases, participants often paused physical actions and visually attended to the agent while awaiting clarification. This pattern was observed across multiple participants, particularly during early stages of task execution.
- **Confirmation of Understanding.** Participants engaged with the agent to confirm their understanding, even when they appeared confident in task execution. These confirmation interactions were typically brief and sometimes self-initiated, serving

as reassurance rather than a request for new information, and were often followed by immediate resumption of the task.

- **Social or Exploratory Engagement.** Some interactions reflected social or exploratory engagement rather than task-oriented needs, such as greeting the agent or responding playfully to its gestures. These interactions occurred primarily at the beginning or end of the session and did not directly influence task progression.
- **Externally Prompted Engagement.** Some interactions were initiated in response to external prompts from the researcher or the agent, and were often characterized by brief or absent visual attention.

We also observed three recurring disengagement patterns in which learners reduced or withdrew attention from the AI agent:

- **Information Redundancy.** Disengagement frequently occurred when agent-provided guidance overlapped with information already available through static interface elements, such as on-screen text panels. In these cases, participants prioritized the text-based information and continued task execution without attending to the agent.
- **Perceived Lack of Conversational Agency.** Disengagement arose when interaction with the agent felt structurally constrained, as the agent's phase-based dialogue limited opportunities for learner-initiated interaction and led participants to treat the agent as a situational support rather than an ongoing conversational partner.
- **Response Latency.** System-related friction such as response latency contributed to disengagement, as participants sometimes resumed task execution before receiving agent feedback and subsequently relied less on the agent for guidance.

## 6 Discussion

Learners generally reported that the overall VR system was easy to use and supportive for learning, reflected in high ratings for interface clarity (e.g., text panels) and embodied interaction (gesture-based manipulation). At the same time, items specific to the AI agent (e.g., ease of communication, learning support, engagement) received lower ratings and showed greater variance. This divergence suggests that learners interpreted the environment as an interface ecology in which static panels served as a low-friction primary channel, while the AI agent was treated as an optional resource. This pattern aligns with immersive learning accounts emphasizing that learning benefits depend not only on immersion but also on how interface elements shape cognitive load and self-regulation [9]. It also resonates with redundancy and split-attention findings in multimedia learning, where duplicative channels can reduce the perceived value of one source—particularly under time or attentional constraints [14].

Our coding results indicate that attention to the AI agent was conditional rather than constant. Across 74 interaction episodes, initiation was most often agent-initiated (32/74; 43.2%), followed by self-initiated (24/74; 32.4%) and researcher-prompted (18/74; 24.3%), and visual attention to the agent was observed in most episodes (64/74; 86.5%). This pattern suggests that learners frequently oriented to the agent at least briefly, but they did not necessarily

sustain or leverage that attention in ways that produced consistently meaningful instructional exchange. In fact, post-survey items specific to the agent received relatively lower and more variable ratings than other system elements, indicating more mixed experiences with the agent despite high overall usability. Consistent with this interpretation, video analysis revealed four recurring motivations for engagement: help-seeking during procedural or conceptual uncertainty, confirmation of understanding, social or exploratory interaction, and compliance with researcher or agent prompting. Interpreted through the initiation  $\times$  attention coding lens, self- and agent-initiated episodes plausibly reflect moments when learners construed the agent as diagnostically useful or socially available, whereas researcher-prompted episodes indicate that the agent's role was not always spontaneously adopted. This pattern is consistent with prior research showing that pedagogical agents in VR do not uniformly improve outcomes; rather, their effectiveness depends on how learners allocate attention and whether agent behaviors align with task context [15]. More broadly, these findings align with cognitive load accounts suggesting that learners are less likely to redirect visual attention during active manipulation [12], as well as with conversational-agent research showing that users shift between functional and social orientations depending on situational demands and interactional context.

Disengagement clustered around three conditions: (1) recognition of redundancy, when agent guidance overlapped with on-screen text; (2) a perceived lack of conversational agency, when the phase-based structure constrained learner-led turns; and (3) response latency, when learners resumed the task before the agent responded. The first condition directly echoes redundancy and split-attention effects in multimedia learning, whereby learners bypass additional sources that introduce coordination costs when equivalent information is available through a faster, lower-friction channel [10]. The second condition aligns with prior work on conversational systems suggesting that effective interaction depends not only on informational accuracy but also on interactional flexibility and user control; when dialogue structures limit brief confirmations or learner-initiated turns, users may reframe the agent as a situational tool rather than a sustained conversational partner [4]. The third condition, response latency, warrants a distinction between disengagement driven by technical limitations and that driven by design choices. Latency-induced disengagement reflects infrastructural constraints of the current pipeline (e.g., network round-trips to the OpenAI Realtime API via LiveKit), suggesting that future implementations may benefit from local or edge-based processing to reduce delays; by contrast, disengagement arising from redundancy or constrained dialogue reflects design-level decisions about information distribution and conversational architecture that require rethinking the agent's role within the interface ecology. This distinction is important because the two sources of disengagement call for fundamentally different optimization strategies—infrastructure improvements in the former case and interaction design revisions in the latter. This interpretation is also consistent with prior findings indicating that delays degrade perceived responsiveness and that lightweight fillers or acknowledgment cues can help preserve interactional continuity [1].

Taken together, RQ1 and RQ2 suggest three design directions for embodied, gesture-performing AI agents in immersive VR. First,

agents should be designed for complementarity within the interface ecology by providing context-sensitive diagnosis, just-in-time hints, and adaptive feedback that static panels cannot offer, avoiding redundancy-driven disengagement [10]. Second, gesture and verbal behaviors should be timed to learners' interaction rhythms, particularly micro-pauses and moments of reduced motor or visual demand, because our observations showed that learners often interacted with the simulation without sustained visual attention to the AI agent. This suggests that placing an AI agent within an immersive simulation does not ensure that the agent is visually attended to during active manipulation. Third, dialogue architectures should preserve phase-based guidance while supporting brief learner-initiated micro-turns (e.g., confirmations, quick checks, short social exchanges) and incorporating acknowledgment cues during latency (e.g., nonverbal fillers) to prevent conversational drop-off and maintain trust in system responsiveness [3].

## 7 Conclusion

This work examined how learners attend to and disengage from a gesture-performing AI agent embedded in an immersive VR learning environment. Through video-based interaction analysis, we found that learners' attention to the agent was conditional rather than persistent, shaped by task context, interface design, and system responsiveness. These findings highlight that merely embedding an AI agent within an immersive simulation is insufficient to secure sustained learner attention. Instead, agent effectiveness depends on how well its behaviors align with learners' embodied interaction rhythms and with the broader interface ecology. We contribute empirical understanding of attention patterns in agent-supported VR environments and design implications for positioning agents as complementary, context-sensitive partners that time gestures to learners' interaction rhythms and support conversational flexibility. Future work will explore agents that leverage multimodal awareness of learners' ongoing activity to dynamically adapt the timing and expressiveness of agent behaviors, better sustaining attention in embodied learning experiences.

## References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233
- [2] Lawrence Barsalou. 2008. Grounded Cognition. *Annual Review of Psychology* 59 (2008), 617–645. doi:10.1146/annurev.psych.59.103006.093639
- [3] Hicham A. Boukaram, Mohamad Ziadde, and Mahmoud F. Sakr. 2021. Mitigating the Effects of Delayed Virtual Agent Response Time Using Conversational Fillers. In *Proceedings of the 9th International Conference on Human-Agent Interaction*. ACM, 130–138. doi:10.1145/3472307.3484171
- [4] Leigh Clark, Nadia Pantidi, Orla Cooney, Philip Doyle, Diego Garaialde, Justin Edwards, Brendan Spillane, Emer Gilmartin, and Benjamin R. Cowan. 2019. What Makes a Good Conversation? Challenges in Designing Truly Conversational Agents. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 1–12. doi:10.1145/3290605.3300705
- [5] Isabel S. Fitton, Daniel J. Finnegan, and Michael J. Proulx. 2020. Immersive virtual environments and embodied agents for e-learning applications. *PeerJ Computer Science* 6 (2020), e315. doi:10.7717/peerj-cs.315
- [6] Arthur Graesser, Patrick Chipman, Brian Haynes, and Andrew Olney. 2005. AutoTutor: An Intelligent Tutoring System with Mixed-Initiative Dialogue. *IEEE Transactions on Education* 48, 4 (2005), 612–618. doi:10.1109/TE.2005.856149
- [7] Kenneth Holstein and Vincent Aleven. 2022. Designing for Human-AI Complementarity in K-12 Education. *AI Magazine* 43, 2 (2022), 239–248. doi:10.1002/aaai.12058
- [8] Lasse Jensen and Flemming Konradsen. 2018. A review of the use of virtual reality head-mounted displays in education and training. *Education and Information Technologies* 23, 4 (2018), 1515–1529. doi:10.1007/s10639-017-9676-0
- [9] Lasse Jensen and Flemming Konradsen. 2018. A Review of the Use of Virtual Reality Head-Mounted Displays in Education and Training. *Education and Information Technologies* 23, 4 (2018), 1515–1529. doi:10.1007/s10639-017-9676-0
- [10] Slava Kalyuga, Paul Chandler, and John Sweller. 2004. When Redundant On-Screen Text in Multimedia Technical Instruction Can Interfere with Learning. *Human Factors* 46, 3 (2004), 567–581. doi:10.1518/hfes.46.3.567.50405
- [11] Todd Kulesza, Simone Stumpf, Margaret Burnett, Sherry Yang, Irwin Kwan, and Weng-Keen Wong. 2013. Too Much, Too Little, or Just Right? Ways Explanations Impact End Users' Mental Models. In *Proceedings of the IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. IEEE. doi:10.1109/VLHCC.2013.6645235
- [12] Guido Makransky and Gustav Bøg Petersen. 2021. The Cognitive Affective Model of Immersive Learning (CAMIL): A Theoretical Research-Based Model of Learning in Immersive Virtual Reality. *Educational Psychology Review* 33, 3 (2021), 937–958. doi:10.1007/s10648-020-09586-2
- [13] Richard E. Mayer. 2002. Multimedia Learning. *Psychology of Learning and Motivation* 41 (2002), 85–139. doi:10.1016/S0079-7421(02)80005-6
- [14] Richard E. Mayer and Roxana Moreno. 2003. Nine Ways to Reduce Cognitive Load in Multimedia Learning. *Educational Psychologist* 38, 1 (2003), 43–52. doi:10.1207/S15326985EP3801\_6
- [15] Gustav B. Petersen, Aske Mottelson, and Guido Makransky. 2021. Pedagogical Agents in Educational VR: An In-the-Wild Study. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, 1–12. doi:10.1145/3411764.3445547
- [16] Gustav Bøg Petersen, Giorgos Petkakis, and Guido Makransky. 2022. A study of how immersion and interactivity drive VR learning. *Computers & Education* 179 (2022), 104429. doi:10.1016/j.compedu.2021.104429
- [17] Jaziar Radianti, Tim A. Majchrzak, Jennifer Fromm, and Isabell Wohlgenannt. 2020. A systematic review of immersive virtual reality applications for higher education: Design elements, lessons learned, and research agenda. *Computers & Education* 147 (2020), 103778. doi:10.1016/j.compedu.2019.103778
- [18] Ido Roll and Ruth Wylie. 2016. Evolution and Revolution in Artificial Intelligence in Education. *International Journal of Artificial Intelligence in Education* 26, 2 (2016), 582–599. doi:10.1007/s40593-016-0110-3
- [19] Margaret Wilson. 2002. Six views of embodied cognition. *Psychonomic Bulletin & Review* 9, 4 (2002), 625–636. doi:10.3758/BF03196322
- [20] Zihao Yi, Jiarui Ouyang, Zhe Xu, Yuwen Liu, Tianhao Liao, Haohao Luo, and Ying Shen. 2025. A Survey on Recent Advances in LLM-Based Multi-turn Dialogue Systems. *Comput. Surveys* 58, 6, Article 148 (2025), 38 pages. doi:10.1145/3771090